

DDM ClickHouse vs PostgreSQL Queries

Query 1

Show all reports where the building type is a Private Home OR Villa, that happened on a weekend (Day of Week = 6 or 7), AND where the call duration was above 30 minutes. Order the results by call duration descending.

SQL

```
SELECT *
FROM noisy_neighbors_reports
WHERE building_type IN ('Home', 'Villa')
      AND Day_of_week IN (6, 7)
      AND Duration_of_Call > 30
ORDER BY Duration_of_Call DESC;
```

Query 2

Within each zip group and day type, keep only the 2 longest calls, breaking ties by earlier report date. Where 'North' means zip 10000-10499, 'Center' means zip 10500-10999, 'South' means zip 11000-11499, 'Weekday' means day of week 1-5, and 'Weekend' means day of week 6 or 7.

SQL

```
WITH classified AS (
  SELECT
    *,
    CASE
      WHEN `Incident Zip` BETWEEN 10000 AND 10499 THEN 'North'
      WHEN `Incident Zip` BETWEEN 10500 AND 10999 THEN 'Center'
      WHEN `Incident Zip` BETWEEN 11000 AND 11499 THEN 'South'
    END AS zip_group,
    CASE
      WHEN `Day of Week` BETWEEN 1 AND 5 THEN 'Weekday'
      ELSE 'Weekend'
    END AS day_type
  FROM noisy_neighbors_reports
  WHERE `Incident Zip` BETWEEN 10000 AND 11499
        AND `Day of Week` BETWEEN 1 AND 7
),

ranked AS (
  SELECT
    *,
    ROW_NUMBER() OVER (
      PARTITION BY zip_group, day_type
      ORDER BY
        `Duration of Call (min)` DESC,
        `Date of Report` ASC
    ) AS rn
  FROM classified
)

SELECT *
FROM ranked
WHERE rn <= 2
ORDER BY
  zip_group,
  day_type,
  `Duration of Call (min)` DESC;
```

Query 3

For each month, compute a monthly metric based on call durations, active report days, and the most frequent borough.

METRIC DEFINITION

$$\text{metric} = \left(\frac{\sum \text{Duration of Call (min)}}{\text{unique report dates in the month}} \right) \times \text{length of the most frequent borough name in that month}$$

SQL

```
WITH base AS (
  SELECT
    *,
    TO_DATE('Date of Report') AS report_date,
    DATE_FORMAT(TO_DATE('Date of Report'), 'yyyy-MM') AS year_month
  FROM noisy_neighbors_reports
),
monthly_stats AS (
  SELECT
    year_month,
    SUM('Duration of Call (min)') AS sum_duration,
    COUNT(DISTINCT report_date) AS uniq_dates
  FROM base
  GROUP BY year_month
),
borough_counts AS (
  SELECT
    year_month,
    'Borough',
    COUNT(*) AS borough_reports
  FROM base
  GROUP BY year_month, 'Borough'
),
max_per_month AS (
  SELECT
    year_month,
    MAX(borough_reports) AS max_borough_reports
  FROM borough_counts
  GROUP BY year_month
),
top_borough_candidates AS (
  SELECT
    bc.year_month,
    bc.'Borough',
    bc.borough_reports
  FROM borough_counts bc
  INNER JOIN max_per_month mp
    ON bc.year_month = mp.year_month
  WHERE bc.borough_reports = mp.max_borough_reports
),
top_borough AS (
  SELECT
    year_month,
    MIN('Borough') AS top_borough,
    LENGTH(MIN('Borough')) AS top_borough_len
  FROM top_borough_candidates
  GROUP BY year_month
)
SELECT
  ms.year_month,
  ms.sum_duration,
  ms.uniq_dates,
  tb.top_borough,
  (ms.sum_duration / ms.uniq_dates) * tb.top_borough_len AS monthly_metric
FROM monthly_stats ms
INNER JOIN top_borough tb
  ON ms.year_month = tb.year_month
ORDER BY ms.year_month ASC;
```